

Unit V – Innovation, Entrepreneurship and Sustainable Design

Detailed lecture notes — innovation and entrepreneurship, value proposition and business models, sustainable and inclusive design, social innovation, the product development cycle, pitching, and ethics in design.

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SYLLABUS

UNIT V — INNOVATION, ENTREPRENEURSHIP AND SUSTAINABLE DESIGN (6 periods)

Introduction to innovation and entrepreneurship. Value proposition. Sustainable and inclusive design. Social innovation. Basics of product development cycle. Pitch preparation and storytelling. Ethical considerations in design.

Course Outcome CO5: Present sustainable and innovative engineering solutions addressing real-world problems.

Text: Tim Brown, *Change by Design*, Harper Collins, 2012; Idris Mootee, *Design Thinking for Strategic Innovation*, 1st Edn., John Wiley & Sons, 2013.

References: Alexander Osterwalder and Yves Pigneur, *Business Model Generation*, Wiley, 2010; Donella H. Meadows, *Thinking in Systems: A Primer*, Sustainability Institute, 2015; Nigel Cross, *Engineering Design Methods*, 4th Edn., Wiley, 2008.

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Learning objectives. After studying this unit you should be able to: (i) distinguish innovation from entrepreneurship and classify an innovation by novelty and by market; (ii) describe the venture journey from problem to scale and the support available in the Indian ecosystem; (iii) write a value proposition and complete a value proposition canvas and a business model canvas for a concept; (iv) apply life-cycle thinking and the principles of a circular economy to a design; (v) explain inclusive and universal design and apply the seven principles; (vi) describe social innovation and the product development cycle with stage-gate review; (vii) prepare and deliver a structured pitch supported by a story; and (viii) identify ethical issues in a design, including dark patterns, and apply a simple test before shipping.

1. Introduction – carrying a solution beyond the classroom

Four units have taken a team from a badly framed problem to a tested prototype. That is where most student projects stop, and where almost all of them die. The prototype works, the report is submitted, the team disperses in April, and the problem the team spent a semester understanding is still there in June. This unit is about the distance between a solution that works and a solution that lasts.

Three things are needed to cross that distance, and they give the unit its shape. The solution needs a **model** — some arrangement by which it is paid for, delivered and maintained after the enthusiasm has gone. It needs to be **sustainable and inclusive** — to survive its own consequences and to work for the people who are usually designed around. And it needs to be **communicated** — because a solution that nobody funds, adopts or permits never reaches anybody at all. Ethics runs through all three, because each of them offers a shortcut that works in the short run and damages somebody.

A prototype proves that a thing can work. A model proves that it can keep working when you are no longer in the room.

2. Innovation and entrepreneurship – two words, two things

The two words are used interchangeably in ordinary speech and mean quite different things. **Innovation** is putting an idea into use so that it creates value. The definition contains two halves and both matter: an idea that is never put into use is an invention or a concept, and a change that creates no value for anybody is novelty, not innovation. Innovation can happen inside an existing organisation, needs no new company and is measured by adoption.

Entrepreneurship is building the vehicle that carries an innovation into the world — assembling people, money and a model, and taking personal risk while doing so. It is measured by survival and by scale. The relationship between the two is asymmetric, and the asymmetry is worth stating plainly to a first-year class: you can innovate without ever starting a company, but you cannot build a lasting company on nothing having been innovated.

	Innovation	Entrepreneurship
What it is	Putting an idea into use so that value is created	Building the organisation that carries it
Where it happens	Anywhere — a company, a college, a government office	Usually in a new venture, sometimes inside a firm
What it needs	An unmet need and a workable idea	A model, money, a team and a tolerance for risk
How it is measured	Adoption — is it being used?	Survival, growth, scale

	Innovation	Entrepreneurship
Its characteristic failure	A clever thing nobody uses	A good product with no way of being paid for

KEY POINT Invention is the first use of an idea. Innovation is its first useful use. Patents record inventions; adoption records innovations, and only the second changes anybody's life.

3. Four types of innovation

A common way of classifying innovation places it on two axes: how new the technology is, and how new the market is. The four quadrants that result behave very differently, need different evidence and fail for different reasons, so a team should know which one it is standing in.

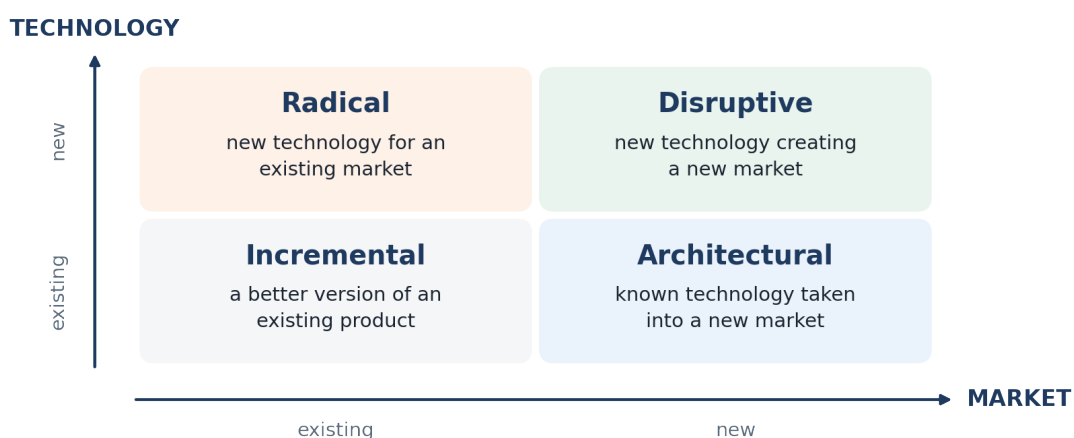


Figure 5.1 Innovation classified by novelty of technology and novelty of market. Risk rises towards the top right; so does the evidence a team must gather.

Type	What is new	Example	Main risk
Incremental	Neither — a better version of an existing product for the existing market	Each successive model of a domestic mixer grinder	Competitors do the same thing
Architectural	The market — known technology taken somewhere it has not been	A clay cooling pot sold as a household refrigerator	The new market behaves unlike the old one
Radical	The technology — a new technology serving an existing market	LED lamps replacing incandescent filament bulbs	The technology may not mature in time, and well-funded incumbents can follow
Disruptive	Both — new technology creating a market that did not exist	A portable, mains-free infant warmer for home births	There is no existing market to measure, so demand is unknown until you try

Two remarks. First, students almost always aim at the top-right quadrant and almost always deliver in the bottom-left, and there is nothing wrong with that: most valuable engineering is incremental, and an incremental innovation that is adopted beats a disruptive one that is not. Second, the quadrants are not a ladder. A team does not graduate from incremental to disruptive; it chooses the quadrant that the need it found actually occupies.

4. What an entrepreneur actually does, and the venture journey

The popular picture of an entrepreneur is a person with a brilliant idea and an appetite for risk. The working reality is duller and more useful: an entrepreneur is somebody who reduces uncertainty faster than they spend money. Almost everything in the early life of a venture is a test — of whether the need is real, whether anyone will pay, whether the thing can be made at a cost that works — and the discipline is the same build-measure-learn loop met in Unit IV, applied to the business rather than to the artefact.

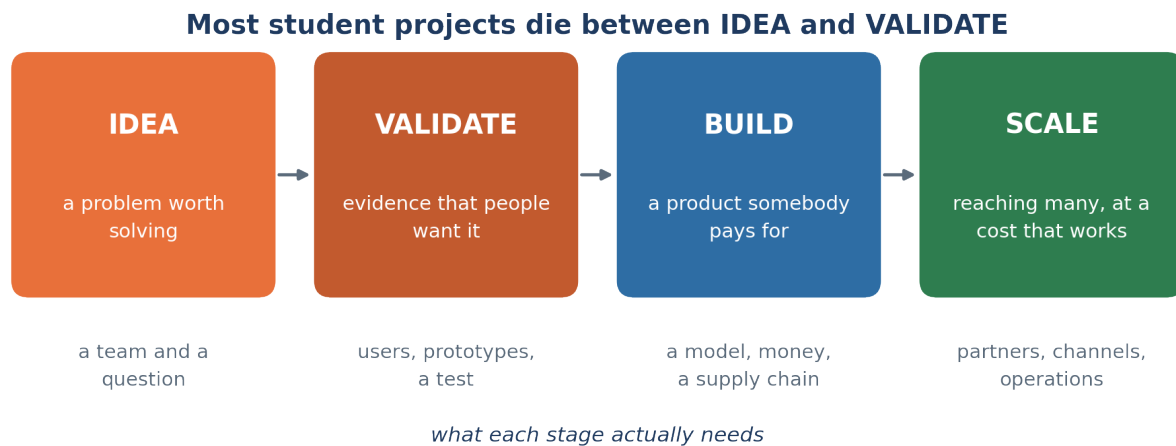


Figure 5.2 The venture journey. Each stage ends with a question answered, and the money needed at each stage is roughly an order of magnitude larger than the last.

1. Problem — a need confirmed in the field, not assumed at a desk. The output is evidence, not an idea.
2. Solution — a prototype tested with users, and the question of whether it actually solves the need answered.
3. Model — how it is paid for, delivered and maintained. The output is a business model canvas that somebody has checked.
4. Pilot — a small real deployment with real users and real money changing hands, usually in one place.
5. Scale — more users, more places, more people in the team, and the problems change character entirely.

The stages are not equally dangerous. Most student ventures — and a great many real ones — die between the second and the third, because the team has a working prototype and no answer to the question of who pays. The second most common death is earlier still: a solution built for a need nobody confirmed.

Why projects die early	What it looks like	The preventive
No real need	Users are polite but never come back	Field evidence from Unit II before anything is built
No model	It works, and there is no way to pay for the next one	A business model canvas written at prototype stage, not after
Founder fatigue	The team graduates and nobody carries it	More than one person who can run it; documentation
Scaling too early	Ten sites opened before one site worked	Make one place work completely first
Ignoring the ecosystem	Everything funded from the team's own pockets	Incubator, grant schemes, competitions — see the next section

KEY POINT An entrepreneur's real skill is not risk-taking but risk-reduction: finding the cheapest experiment that removes the largest uncertainty, and running it before the money is committed.

5. The ecosystem – what is available to a student in India

A student team rarely fails for want of an idea. It fails for want of the boring things — a place to work, a little money, somebody who has done it before, and somebody who will be the first customer. A considerable ecosystem exists in India to supply exactly those, and most first-year students are unaware of it.

Where support comes from	What it offers	How a student reaches it
The college itself	Incubation or innovation cell, laboratories, workshop time, faculty mentors, project funding, the IEDC or entrepreneurship cell	Ask the department; this is always the first door
Atal Innovation Mission (NITI Aayog)	Atal Tinkering Labs in schools and Atal Incubation Centres, with grants and mentorship for early ventures	Through an affiliated incubation centre
Startup India (DPIIT)	Recognition for startups, with associated tax and procedural benefits, and a public learning and networking platform	Register the venture once it is a legal entity
State government	State startup policies and missions — in Tamil Nadu, StartupTN — offering seed grants, co-working space and events	State startup portal and district-level events
National agencies	DST, BIRAC, MSME and similar bodies funding technology and social ventures, often through host institutions	Almost always applied for through an institution, not individually
Competitions and hackathons	Smart India Hackathon, institutional expos, industry challenges — money, visibility and a deadline	Watch departmental notice boards; entry is usually free

KEY POINT The first resource on that list is the one students forget. A faculty mentor and a college incubation cell cost nothing, are available in the first week of a project, and are the route to almost everything else on the list.

Two cautions about the ecosystem. First, do not chase funding before you have evidence: money arriving before the need is confirmed buys a larger version of the wrong thing. Second, a competition is not a customer. Winning a prize proves that a panel liked the pitch; it does not prove that anybody will use the product, and a team that mistakes the first for the second stops testing at exactly the wrong moment.

The third route out of the classroom is often overlooked. **Intrapreneurship** is entrepreneurial work done inside an existing organisation — improving a process, launching an internal product, changing how a department serves the people it serves. It carries far less personal risk, it uses the organisation's resources and distribution, and it is what most graduates of this course will actually do. The methods of this course apply unchanged; only the vehicle is different.

6. Value proposition and the value proposition canvas

A **value proposition** is a plain statement of what a user gains, for whom, and why that is better than what they do today. It is the hinge of the whole unit: everything before it exists to discover it, and everything after it — the business model, the pitch, the funding — is built on it. Teams write it badly, and they do so in a predictable way, by describing the product instead of the gain.

Weak	Why it is weak	Stronger
"An IoT-based smart water management solution"	Describes technology; names no user and no gain	"Hostel residents stop losing water to overflowing tanks, without anyone having to watch the tank"
"A mobile app for campus issues"	Any app, any campus, any issue	"A student can report a broken tap in fifteen seconds and see when it was fixed"
"A low-cost affordable device for farmers"	"Low-cost" is a price, not a value; "farmers" is not a user	"A small-holding paddy farmer can test soil moisture herself instead of waiting three days for the lab"

A serviceable template, filled from the team's own field notes rather than invented: **For** [a specific user] **who** [has this situation], **our** [solution] **gives** [this gain] **unlike** [what they do today], **because** [the reason it is different]. If any bracket cannot be filled from evidence, the team has found its next piece of work.

6.1 The value proposition canvas

The **value proposition canvas**, devised by **Alexander Osterwalder** and colleagues and published in *Value Proposition Design* in 2014, makes the fit between a user and an offer explicit. It has two halves that must be filled in separately and then compared: a circle for the customer and a square for the offer.

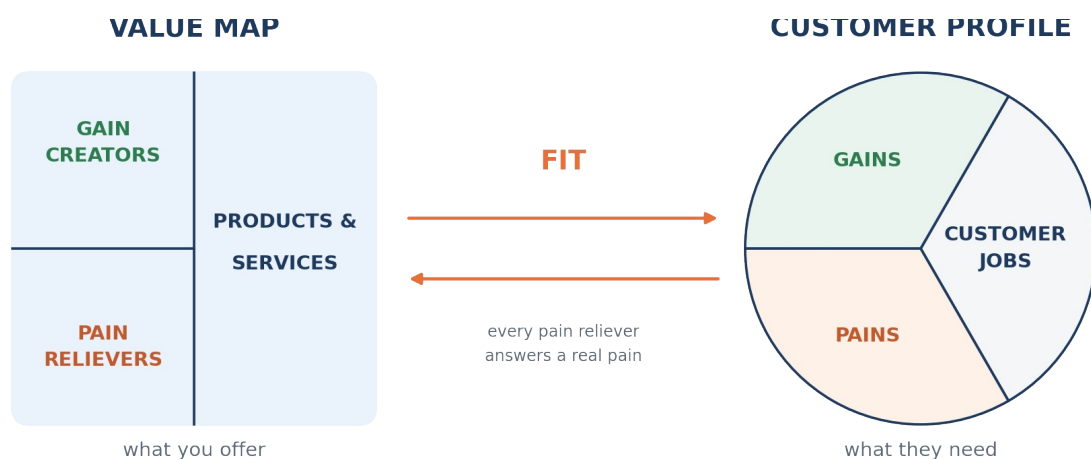


Figure 5.3 The value proposition canvas. The customer side is filled from field evidence first; the offer side is then designed to fit it, never the other way round.

Customer side (the circle)	Offer side (the square)
Jobs — what the user is trying to get done, functional, social and emotional	Products and services — what you actually give them
Pains — what is annoying, costly, risky or embarrassing about doing it today	Pain relievers — specifically how your offer removes those pains
Gains — what a good outcome looks like, including gains they do not expect	Gain creators — how your offer produces those gains

Fill the circle before the square. A canvas whose customer side was written to justify the product already on the square is a decorated opinion.

The canvas is read for **fit**: every pain reliever should point at a pain that was actually recorded in the field, and every important pain should have something pointing at it. Two failure patterns show up immediately. Pain relievers with no matching pain are features the team wanted to build. Pains with nothing pointing at them are either the next piece of design work or an honest admission that this concept does not serve them.

IN CLASS Twenty-five minutes. Complete a value proposition canvas for your concept, filling the customer side only from your Unit II field notes. Draw a line from every pain reliever to the pain it addresses, and circle any pain reliever with no line and any pain with no line. Write your value proposition in the template above.

7. The business model canvas

A business model is the description of how an organisation creates, delivers and captures value. The **business model canvas** of **Osterwalder and Pigneur**, published in *Business Model Generation* in 2010, puts it on a single sheet with nine blocks, and it has become the standard instrument for the purpose. For a student team its virtue is that it makes the missing block obvious: the typical project canvas is rich on the left and empty under revenue, which is precisely the gap that kills such projects.

Nine blocks — Osterwalder, 2005



Figure 5.4 The nine blocks of the business model canvas. The right-hand side concerns the customer, the left-hand side concerns the operation, and the two meet at value.

Block	The question it answers	A student example
Customer segments	Whose problem is this, specifically?	First-year hostel residents in two blocks
Value proposition	What gain do they get, and why is it better?	A fault is reported in fifteen seconds and its status is visible
Channels	How does it reach them?	A sticker with a code on each fitting; the hostel notice board
Customer relationships	How is the relationship maintained?	Self-service, with a monthly report to the warden
Revenue streams	Who pays, how much, and for what?	Nobody pays; the hostel office funds it from its maintenance budget

Block	The question it answers	A student example
Key resources	What do we need to have?	A hosted list, the stickers, one student administrator
Key activities	What must we do, repeatedly?	Keep the list current, close reported faults, publish the report
Key partners	Who do we need beside us?	The hostel office, the plumbing contractor, the college IT cell
Cost structure	What does it cost to run for a year?	Hosting, printing, and the time of one student

Three rules make the canvas useful rather than decorative. Fill it in **pencil** — every block is a hypothesis and most will change. Fill the **revenue and cost blocks honestly**; a project whose answer is “the college pays” is perfectly respectable, but that answer must be written down and checked with the college rather than assumed. And distinguish the **user** from the **payer**: in public services, healthcare and campus systems they are routinely different people, and a model that confuses them fails at the pilot stage.

KEY POINT The block a student canvas leaves blank is almost always revenue, and the blank is not a formatting problem. If nobody pays and nothing funds the running cost, the solution ends on the day the team graduates.

IN CLASS Thirty minutes. Complete all nine blocks of a business model canvas for your concept. Name a real payer for the revenue block and estimate a real annual figure for the cost block. Identify the block you are least certain about and state the cheapest experiment that would reduce that uncertainty.

8. Sustainable design and the life-cycle view

A design does not stop existing when it is sold. It is made from materials that came from somewhere, it is transported, used for some years, maintained or not, and then disposed of by somebody. Sustainable design is the practice of taking that whole span into account while the design is still on paper, because almost all of the environmental consequence of a product is locked in by decisions taken in the first few weeks of designing it.

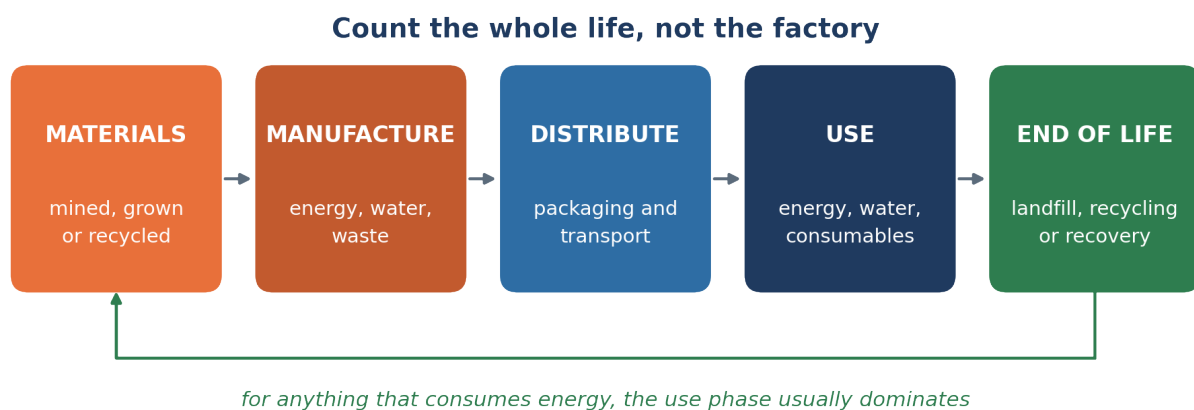


Figure 5.5 The life-cycle stages of a product. The design decision is taken at the start; the consequences arise at every later stage, and mostly to other people.

Life-cycle stage	What to ask at the design stage
Raw material	Where does it come from? How much is needed? Is there a less damaging material that would do?
Manufacture	How much energy and water? How many separate processes? How much is scrapped?
Distribution	How far does it travel? Does it pack efficiently? Can it be made nearer the user?
Use	What does it consume in service? How long does it last? Can it be repaired by its owner?
End of life	Who disposes of it, and how? Can it be separated into materials? Is anything in it hazardous?

A full life-cycle assessment is a specialist quantitative exercise and is beyond this course. What is not beyond it — and is expected in the project — is **life-cycle thinking**: asking the five questions above, answering them qualitatively, and letting the answers change the design. A student team that chooses a screwed joint over a glued one so that the housing can be opened and repaired has done sustainable design, without measuring a single kilogram of carbon.

- **Design for long life before anything else.** A product that lasts twice as long halves nearly everything, and no recycling scheme matches that.
- **Design for repair.** Standard fasteners, accessible parts, published spares. A glued-shut housing is a disposal decision taken at the design stage.
- **Use fewer materials, and fewer kinds.** A part made of one polymer can be recycled; the same part made of three bonded together cannot.
- **Reduce what it consumes in service.** For most powered products the use phase dominates everything else.
- **Do not design for disposal by somebody poorer than you.** Electronic waste that leaves a city is dismantled by hand, by somebody, somewhere.

Roughly four-fifths of a product's environmental impact is committed by the end of the design stage, when almost nothing has yet been spent. That is the whole argument for thinking about it now.

9. From a linear to a circular economy

The industrial model inherited from the twentieth century is linear: take materials, make a product, use it, throw it away. It works only as long as materials are limitless at one end and space is limitless at the other, and neither is true. A **circular economy** — a term brought into wide use by the **Ellen MacArthur Foundation** from 2010 — designs the end of life back into the beginning, so that materials and products circulate rather than leave.

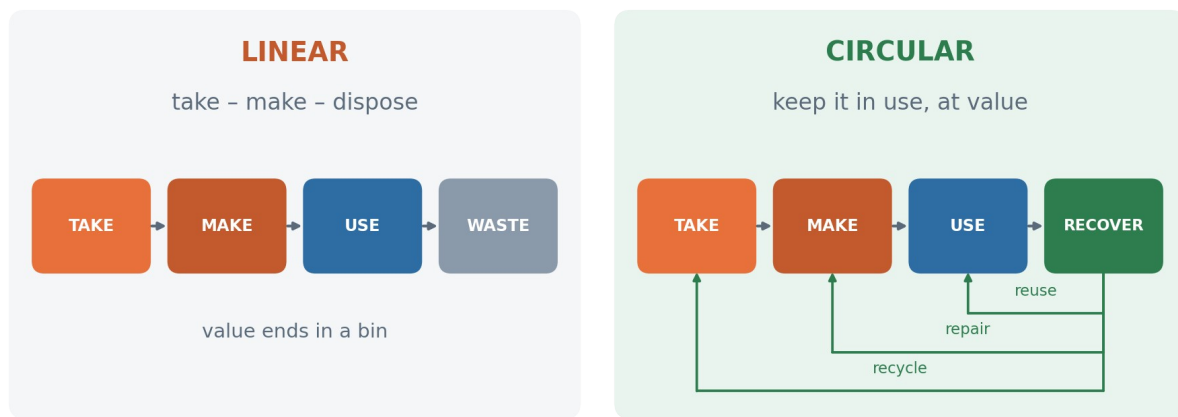


Figure 5.6 Linear and circular models. In the circular model every arrow that returns is a design decision taken long before the product was made.

Strategy	What it means	What the designer must do
Refuse / rethink	Question whether the product is needed at all	Ask whether a service, a shared object or a repair would serve better
Reduce	Less material, less energy, less packaging	Simplify the part count and the material list
Reuse	The same object used again by somebody else	Design for durability and for a second owner
Repair	Faults fixed rather than the product replaced	Standard fasteners, available spares, published instructions
Refurbish / remanufacture	Returned products rebuilt to as-new condition	Design modules that can be swapped; build a return route
Recycle	Materials recovered as materials	Single materials, easy separation, no mixed laminates
Recover	Energy recovered when nothing else is possible	The last resort, not a plan

KEY POINT The strategies are in order of value. Recycling is the second-worst thing that can happen to a product, and a design whose only environmental claim is that it is recyclable has skipped six better options.

The circular idea also changes business models, and this is where it touches the previous section. If a company keeps ownership and sells the use — lighting rather than light fittings, a service contract rather than a pump — then durability and repairability become sources of profit instead of costs. A model of that kind is within reach of a student project: a hostel that hires out cycles has a much stronger reason to maintain them than a hostel that sells them.

10. Inclusive and universal design

Designs are usually made for an imagined average person, and that person does not exist. Real users differ in height, strength, sight, hearing, dexterity, language, literacy and familiarity with technology, and they differ across their own lives — the same person is temporarily one-handed while carrying a child and permanently one-handed after an accident. **Inclusive design** is the practice of designing so that as many people as possible can use the result without adaptation.

Universal design is the term coined by the architect **Ronald Mace**, himself a wheelchair user, and the seven principles below were formulated by a group he led at North Carolina State University in 1997. They are written for the built environment but apply without modification to products, interfaces and services.

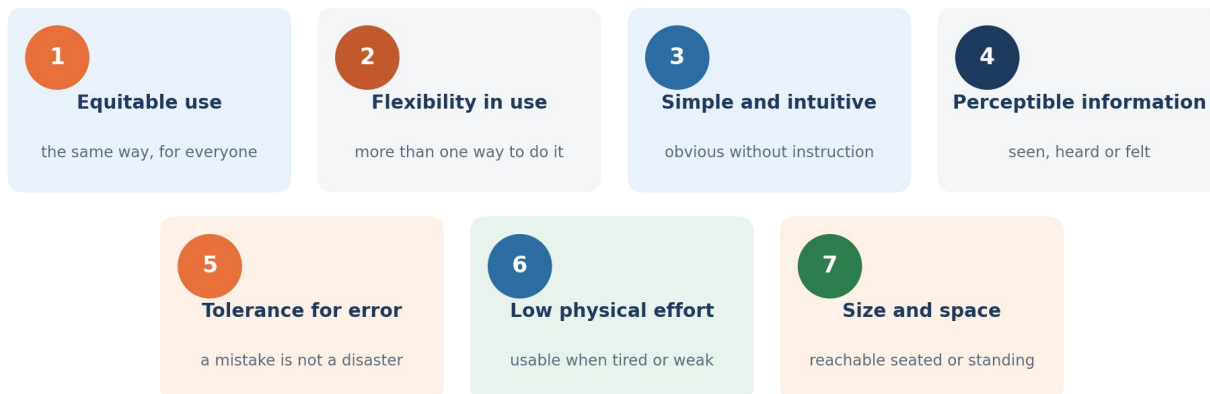


Figure 5.7 The seven principles of universal design. Each is a question to put to a design, not a box to tick after it is finished.

Principle	What it requires	Example
Equitable use	The same means for everyone where possible, never segregating	A level entrance used by all, rather than a ramp at the back
Flexibility in use	Accommodates a range of preferences and abilities	Scissors usable in either hand
Simple and intuitive	Understandable regardless of experience or language	A tap whose direction of turn is obvious
Perceptible information	Conveyed in more than one mode	A lift that announces the floor and displays it
Tolerance for error	Mistakes are recoverable and not punished	An undo; a plug that cannot be inserted the wrong way
Low physical effort	Usable without fatigue	A lever handle rather than a round knob
Size and space for approach	Reachable whatever the user's body or posture	Controls at a height reachable seated or standing

Design for the extremes and the middle is included. Design for the average and the extremes are excluded — and there are more people at the extremes than anybody expects.

The practical argument is the **curb-cut effect**: the dropped kerb was built for wheelchairs and is used far more by people with trolleys, suitcases, prams and bicycles. Captions were made for deaf viewers and are used by everybody in a noisy room. Accessibility features are almost never used only by the group they were made for, which is why inclusive design is a design argument and not only an ethical one.

IN CLASS Twenty minutes. Test your concept against the seven principles, recording a specific pass or fail with a reason for each. Then choose one failure and redesign for it. Note whether the redesign made the concept worse for anybody else — if it did, you have found an adaptation rather than an inclusive design.

11. Social innovation

Social innovation is innovation whose primary purpose is to meet a social need — health, education, water, sanitation, livelihood, mobility — rather than to capture a market. It is not charity, and the distinction matters: charity gives away a thing, while social innovation changes the arrangement that produced the need, and is expected to sustain itself financially in some way.

	Commercial innovation	Social innovation
Primary aim	Profit and growth	A social outcome that lasts
Who is served	Those who can pay	Often those who cannot pay the full cost
Revenue	From the user	Cross-subsidy, government, philanthropy, or a mix
Measured by	Revenue, market share	Outcomes: cases treated, children in school, litres delivered
Characteristic risk	Competition	Dependence on a grant that ends

The Unit I case studies are all social innovations, and re-reading them with this unit's vocabulary is instructive. The **Aravind Eye Care System** does not give surgery away; it charges paying patients enough to cover those who pay nothing, and drove its own cost down far enough for the arithmetic to work — that is a business model, not a donation. The **Jaipur Foot** is designed around squatting, walking on uneven ground and working without shoes, and is fitted the same day — a design decision, an operational decision and a funding decision that together make it reach people. **Embrace** reached infants no incubator could reach because it needed no continuous electricity, and its story began with a field observation that overturned the team's own brief.

KEY POINT A social innovation without a model is a pilot that ends when the grant does. The question “who pays for the second year?” is not a commercial intrusion into social work; it is the difference between a project and an institution.

12. The product development cycle and stage-gate review

Design thinking is iterative; an organisation that has to commit money and people also needs a way of deciding when to commit more. The two are reconciled by a **stage-gate process**, formalised by **Robert G. Cooper** in the 1980s, in which the work is divided into stages and each stage ends at a gate where a decision is taken on evidence. Inside a stage the team iterates freely; at a gate, somebody decides.

Stages do the work. Gates decide whether it continues.

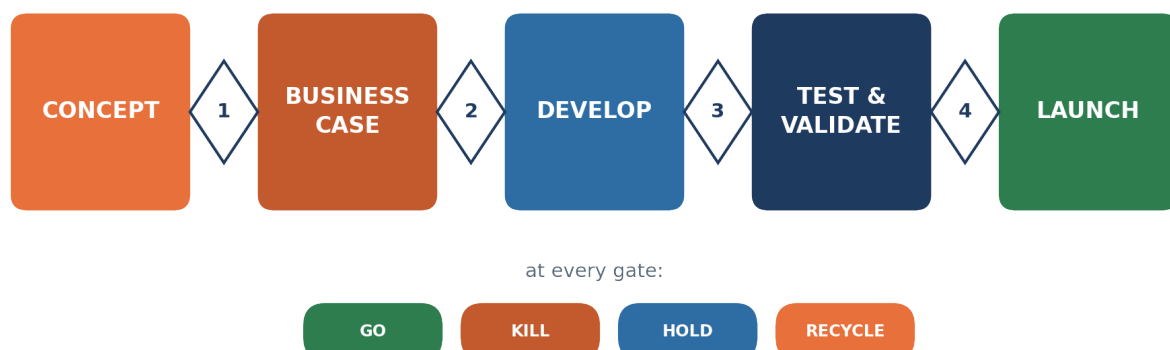


Figure 5.8 Stages and gates. Iteration happens inside the stages; the gates are where money is committed and where a project may be stopped.

The figure shows the process compressed into five stages, which is how it is usually drawn on a wall. Cooper's full description separates the earliest work into discovery and scoping, and the table below expands it that way.

Stage	What happens	The gate question
Discovery	Field work, framing, opportunity identified	Is there a real need, and is it ours to work on?
Scoping	Concepts generated and screened, rough feasibility	Is there a concept worth developing?
Business case	Value proposition, model, cost and benefit	Does it make sense to build — technically and financially?
Development	Design, prototypes, iterations, testing	Does it work, and does it meet the requirement?
Testing and validation	Pilot with real users in real conditions	Does it work outside the laboratory?
Launch	Manufacture, deploy, support	Is the organisation ready to sustain it?

The value of a gate lies in its being allowed to say no. A gate that has never stopped a project is a ceremony, and the projects it waves through consume the resources that the good ones needed. Three decisions are available at any gate — **go**, **kill** and **hold or recycle** — and the third, sending the work back for more evidence, is the most common honest answer.

KEY POINT Stopping a project at a gate is a success of the process, not a failure of the team. The alternative is to discover the same thing after tooling, when it costs a hundred times as much to act on.

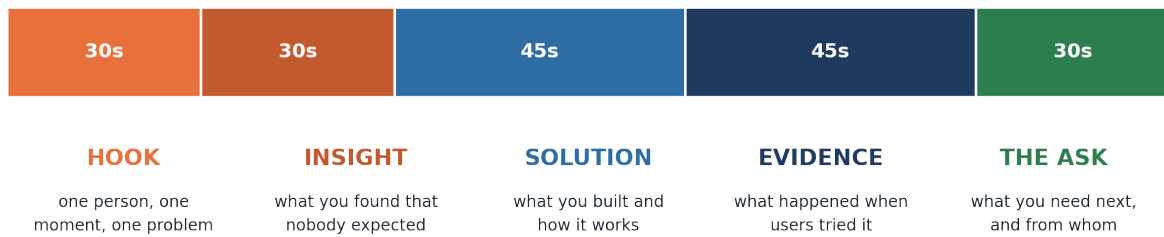
In this course the gates are the reviews. Your problem statement at the end of Unit II, your selected concept at the end of Unit III, your tested prototype at the end of Unit IV and your pitch at the end of Unit V are four gates, and at each one the question is the same: what is the evidence, and what would change your mind?

13. Pitch preparation and storytelling

A pitch is a short, structured case for a decision. It is not a summary of the semester and it is not a demonstration of effort. Somebody in the room is being asked to do something — fund it, permit it, adopt it, join it — and everything in the pitch either moves them towards that or wastes their attention.

The reason storytelling is taught alongside it is not decoration. A list of features is forgotten within minutes; a specific person in a specific difficulty is remembered, and the audience carries that person into the discussion that happens after the team has left the room. The material for the story already exists in the team's field notes, and the temptation to invent a more dramatic user should be resisted, because invented users do not survive questions.

Three minutes, five moves



open with a person, close with a request — never with “thank you, any questions?”

Figure 5.9 The arc of a pitch. Attention is highest at the opening and at the ask; the middle is where evidence goes.

1. Open with one person, named, in thirty seconds. Not the market, not the statistics — one person your team actually met.
2. State the problem as they experience it, and only then widen it to how many others share it.
3. Show the insight: the thing you learnt in the field that the audience does not know. This is the moment a pitch either earns attention or loses it.
4. Present the solution in one sentence before any detail. If the audience cannot repeat it, no amount of detail will help.
5. Show the evidence: the prototype, what you tested, with how many users, and what they did — including what went wrong.
6. Explain the model briefly: who pays, what it costs, who delivers it.
7. Make the ask, specifically. “We need workshop access and two thousand rupees to test with thirty users” is an ask; “we hope to take this forward” is not.

Do	Do not
Start with a person and a situation	Start with “India has 1.4 billion people”
Say the solution in one sentence, early	Withhold it for suspense
Show the prototype, or a photograph of it in use	Show a rendering and call it a prototype
Report what failed in testing and what you changed	Report only successes; experienced listeners distrust them
Give real numbers with their source	Give impressive numbers with no source
Rehearse aloud and time it	Read the slides to the audience
End with a specific ask	End with “thank you, any questions?”

If the audience remembers only one sentence, which sentence do you want it to be? Write it down, and then build the pitch so that it is unmissable.

14. The pitch deck and its delivery

The conventional structure of a short pitch deck is well established, and there is nothing to be gained from departing from it. Ten slides is a good target for a five-minute pitch, and the discipline it imposes — roughly thirty seconds a slide — is itself useful.

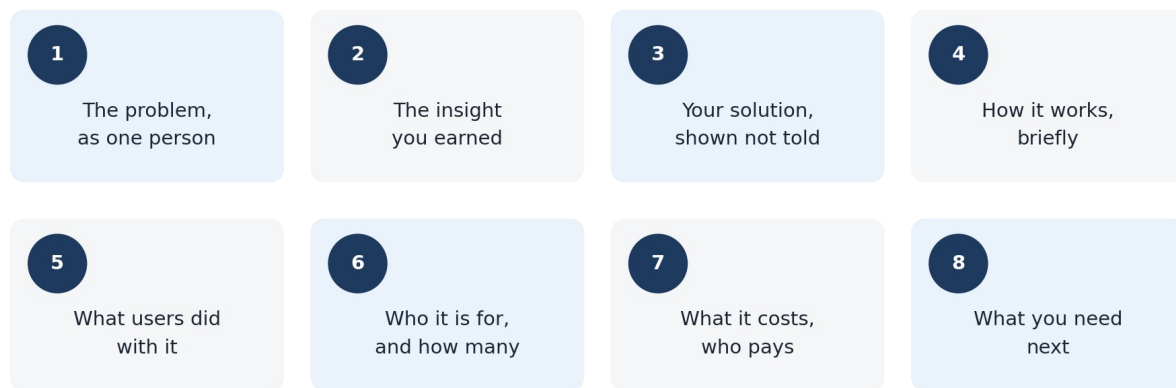


Figure 5.10 A ten-slide structure for a five-minute pitch. The order is conventional because an audience knows it and can follow without effort.

Slide	Content	Common student error
1. Title	Project name, team, one line on what it is	A decorative title with no idea of what the project does
2. The person	One user, named, with a photograph if permitted	A stock photograph of nobody in particular
3. The problem	What goes wrong today, shown rather than asserted	A paragraph of text read aloud
4. The insight	What you learnt in the field that is not obvious	Skipped entirely — and it is the most valuable slide
5. The solution	One sentence and one image	Six features and a block diagram
6. How it works	The mechanism or the flow, simply	A circuit diagram nobody in the room can read
7. Evidence	Testing: how many users, what they did, what changed	“Users liked it”
8. The model	Who pays, what it costs to run	A five-year revenue projection invented in the hostel
9. What is next	The next experiment and the next gate	A product roadmap to 2035
10. The ask	Exactly what you need, from this audience	“Thank you”

- **One idea per slide.** If a slide needs a paragraph, it is two slides or it is something you say aloud.
- **Large type, few words.** The audience reads or listens; it does not do both. Anything you intend to say should not also be written.
- **Show the real thing.** A photograph of a user holding your rough prototype is worth more than any rendering, and it proves the test happened.
- **Rehearse standing up, aloud, with a timer.** Silent rehearsal always runs about thirty per cent faster than the real thing.
- **Prepare for three questions.** Who pays? What is the evidence? What happens when it fails? Every panel asks at least two of them.
- **Answer “we do not know yet” honestly.** Followed by how you would find out, it is a strong answer. An invented number is fatal, because the one person who knows the real figure is usually in the room.

IN CLASS Forty minutes. Build a ten-slide deck to the structure above and rehearse a five-minute pitch aloud, timed. Present to another team, who must then state your value proposition in their own words and ask you the three standard questions. Record where they were unable to repeat what you said.

15. Ethical considerations in design

Every decision in this course has been a choice about somebody else's life. Whose problem was worked on, who was interviewed and who was not, what the prototype made easy and what it made hard, who pays and who is excluded — all of these are ethical choices, and they are made whether or not anybody notices making them. Ethics in design is not a lecture appended at the end; it is a description of what the team has been doing all semester.

15.1 Where the choices arise

Where	The question	The failure it prevents
Choosing the problem	Whose problem are we solving, and who decided?	Solving a comfortable problem for people who already have options
Field work	Did they consent, do they know what we will do with it, can they stop?	Treating people as a source of data
Framing	Whose account of the problem is in our statement?	The powerful stakeholder's version becoming the official one
Ideation and selection	Who benefits, who bears the cost, who is left out?	A solution that works by shifting the burden elsewhere
Prototyping and testing	Is anybody at risk in the test? Was the wizard revealed?	Harm during testing; undisclosed deception
Data	What do we collect, why, for how long, and who can see it?	Collecting everything because it was easy to collect
Business model	Does it make money from a user's confusion or weakness?	A model that requires people to make mistakes
Sustainability	Who lives with the waste?	Exporting the consequence to somebody poorer

15.2 Dark patterns

A **dark pattern** is an interface deliberately designed so that a user does something they did not intend. The term was coined by **Harry Brignull** in 2010. These are not accidents or the result of poor skill; each one takes deliberate design effort, which is precisely what makes them an ethical matter rather than a usability defect. They are included in this course because a student who cannot name them will reproduce them without noticing.



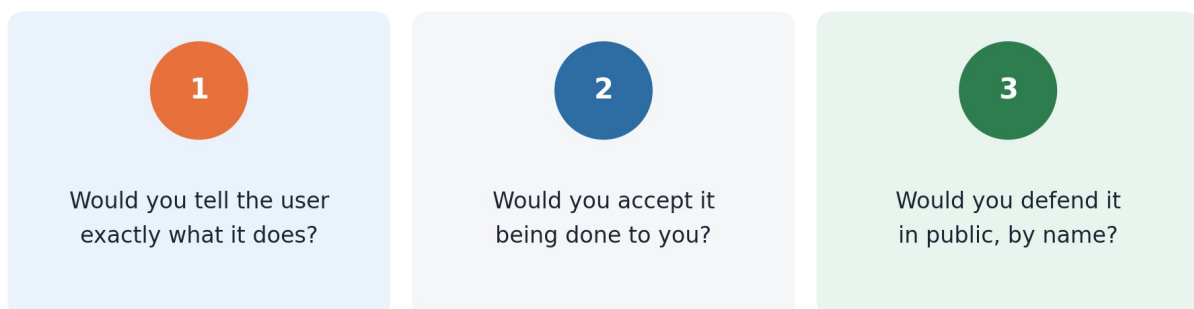
Figure 5.11 Common dark patterns. Each requires deliberate design work, which is what distinguishes a dark pattern from a usability fault.

Pattern	What it does
Roach motel	Easy to sign up, deliberately hard to leave
Confirmshaming	Makes the decline option humiliating — “No thanks, I don’t want to save money”
Misdirection	Draws the eye to the choice that benefits the company
Sneak into basket	Adds something the user did not choose
Drip pricing	Fees revealed only at the last step, after the user is committed
Forced continuity	A free trial that silently becomes a paid subscription
Privacy by default	Sharing is switched on unless the user goes and finds it
Disguised advertisement	An advertisement dressed as content or as a control
Nagging	Repeated requests until the user gives in to stop being asked

If the design works only because the user misunderstands it, it is not a clever design. It is a design that depends on the user's mistake.

15.3 A test before you ship

Three questions before you ship



any “no” is the design telling you something

Figure 5.12 Three questions to put to a design before it reaches anybody. A design that fails any one of them needs changing, not defending.

1. **Transparency.** Would you tell the user exactly what this does, in full and in plain words? A design that needs concealment has already failed.
2. **Reversibility.** Would you accept this being done to you — and specifically to you in the weakest position, not the strongest?
3. **Publicity.** Would you defend this in public, by name, if it were described accurately in a newspaper?

These are informal versions of long-standing ethical tests, and their virtue is that they take under a minute and are hard to argue with. A fourth is worth adding when a decision looks small: if everybody in your position designed this way, would the result be a world you would want to live in? Small dishonesties pass the first three tests one at a time and fail this one together.

15.4 The engineer's professional obligation

The obligation is not merely personal. The code of ethics of the **Institution of Engineers (India)** requires, among other things, that an engineer hold paramount the safety, health and welfare of the public, practise only in areas of competence, issue statements in an objective and truthful manner, disclose conflicts of interest, and protect the environment for sustainable development. Every one of these bears directly on a design project, and a student who has made choices about who is served, what is claimed and what is discarded has already been practising — or failing to practise — that code.

KEY POINT The ethical question is almost never “is this legal?”. It is “who is worse off because of this, and did they have any say?” A team that can answer both questions about its own project has done the work of this section.

IN CLASS Twenty minutes. Put your own concept through the questions above and record the answers. Identify any dark pattern your interface or your model relies on, even mildly. Name one group who is worse off because of your design and state what you would change if you had another two weeks.

16. Summary

- A prototype shows that a thing can work; a model shows that it keeps working after the team has gone. Crossing that distance needs a model, sustainability and communication, with ethics running through all three.
- Innovation is putting an idea into use so that value is created; entrepreneurship is building the vehicle that carries it. You can innovate without a company, but not build a lasting company without innovation.
- Innovation is classified by novelty of technology and of market into incremental, architectural, disruptive and radical; most valuable engineering is incremental, and an adopted incremental beats an unadopted disruptive one.
- An entrepreneur's real skill is reducing uncertainty faster than money is spent; the venture journey runs problem, solution, model, pilot, scale.
- Most student ventures die between solution and model, for want of an answer to who pays; the next most common death is a solution to an unconfirmed need.
- The ecosystem begins with the college's own incubation cell and mentors, and extends through Atal Innovation Mission, Startup India, state missions such as StartupTN, national agencies and competitions.
- A competition is not a customer, and funding that arrives before evidence buys a larger version of the wrong thing.

- A value proposition names a specific user, the gain, and why it beats what they do today; it describes the gain, never the technology.
- The value proposition canvas pairs customer jobs, pains and gains with products, pain relievers and gain creators; the circle is filled from field evidence before the square is designed.
- The business model canvas holds nine blocks; the block student teams leave blank is revenue, and the user and the payer are frequently different people.
- Roughly four-fifths of a product's environmental impact is committed during design; life-cycle thinking asks about material, manufacture, distribution, use and end of life while change is still free.
- Circular strategies run refuse, reduce, reuse, repair, refurbish, recycle, recover, in that order of value; recycling is near the bottom, not the top.
- Inclusive design serves the widest range of people without adaptation; the seven principles of universal design come from Ronald Mace's group at NC State, 1997.
- Designing for the extremes includes the middle, and accessibility features are used far beyond the group they were made for — the curb-cut effect.
- Social innovation meets a social need and is not charity: it needs a model, and the question “who pays for the second year?” separates a project from an institution.
- A stage-gate process allows free iteration within stages and a decision on evidence at each gate; a gate that never stops a project is a ceremony.
- A pitch is a case for a decision, not a summary of effort: one named person, the problem, the insight, the solution in a sentence, the evidence, the model, and a specific ask.
- A ten-slide deck at roughly thirty seconds a slide, rehearsed aloud and timed, with the prototype shown in use and failures reported honestly.
- Ethical choices are made throughout the process, not at the end; dark patterns are deliberate design work, and a design that needs concealment has already failed the publicity test.
- The Institution of Engineers (India) code requires paramount regard for public safety, health and welfare, competence, truthfulness, disclosure of conflicts and protection of the environment.

17. Review questions

17.1 Short-answer (two-mark) questions

1. Distinguish innovation from entrepreneurship.
2. Distinguish invention from innovation.
3. Name the two axes of the innovation matrix and the four resulting types.
4. Give one example each of an incremental and a disruptive innovation.
5. State the five stages of the venture journey.
6. Give two reasons student ventures commonly die early, with the preventive for each.
7. What is intrapreneurship? Why does it matter to most graduates?
8. Name four sources of support available to a student venture in India.
9. Why is a competition not a customer?
10. Define a value proposition and state its three required components.
11. Rewrite “an IoT-based smart irrigation solution” as a value proposition.
12. Name the three elements on each side of the value proposition canvas.
13. What does “fit” mean on a value proposition canvas?
14. Name the nine blocks of the business model canvas.
15. Why must user and payer be distinguished in a business model?
16. List the five life-cycle stages of a product.

17. Why is design for long life ranked above recycling?
18. State the circular-economy strategies in order of value.
19. Distinguish inclusive design from an adaptation for a particular group.
20. State any four of the seven principles of universal design.
21. Explain the curb-cut effect with an example.
22. Distinguish social innovation from charity.
23. What is a stage gate? Name the three decisions available at one.
24. Why is a gate that has never stopped a project a problem?
25. State the seven parts of a pitch in order.
26. What makes an ask specific? Give one example and one counter-example.
27. Define a dark pattern and name three examples.
28. State the three questions of the ethics test before shipping.
29. State any four obligations in the Institution of Engineers (India) code of ethics.

17.2 Long-answer (thirteen-mark) questions

1. Distinguish innovation from entrepreneurship and from invention. Explain the four types of innovation with examples, and discuss the characteristic risk of each.
2. Describe the venture journey from problem to scale. Explain what question each stage answers and discuss, with reasons, the two stages between which student ventures most commonly fail.
3. Discuss the innovation ecosystem available to a student in India. Describe at least five sources of support, what each provides and how it is reached, and state two cautions about using them.
4. Explain what a value proposition is and how it is written. Analyse three weak statements, explain why each fails, and rewrite them.
5. Explain the value proposition canvas in full. Describe both sides, the meaning of fit, and the two failure patterns visible on a completed canvas.
6. Explain the nine blocks of the business model canvas with a worked example from a campus or community project. Discuss why the revenue block is usually the weakest.
7. Explain life-cycle thinking. Describe the five stages, the question to ask at each during design, and five strategies a designer can apply, with examples.
8. Compare the linear and circular economic models. Explain the circular strategies in order of value and discuss how a circular approach changes a business model.
9. Explain inclusive and universal design. State and illustrate the seven principles, and discuss the curb-cut effect with at least two examples.
10. Explain social innovation and distinguish it from both charity and commercial innovation. Analyse two Indian examples in terms of their model, not only their product.
11. Describe the product development cycle with stage-gate review. Explain the stages, the gate questions and the three gate decisions, and discuss how iteration and gating are reconciled.
12. Explain how a pitch is prepared and delivered. Describe the seven-part structure, the role of storytelling, and the three questions every panel asks.
13. Describe a ten-slide pitch deck. State the content of each slide, the common student error on each, and the rules for slide design and rehearsal.
14. Discuss ethical considerations across the design process. Identify the choice arising at each stage and the failure it prevents, and explain the transparency, reversibility and publicity tests applied before shipping.
15. Explain dark patterns with at least five named examples. Discuss why they are an ethical rather than a usability matter, and relate them to the engineer's professional code of ethics.

Text: Tim Brown, Change by Design, Harper Collins, 2012 — the chapters on new social contracts and design activism; Idris Mootee, Design Thinking for Strategic Innovation, Wiley, 2013 — Chapters 7 and 8. **References:** Alexander Osterwalder and Yves Pigneur, Business Model Generation, Wiley, 2010; Alexander Osterwalder et al., Value Proposition Design, Wiley, 2014; Donella H. Meadows, Thinking in Systems: A Primer, Sustainability Institute, 2015; Ellen MacArthur Foundation, Towards the Circular Economy, 2013; Molly Follette Story, James L. Mueller and Ronald L. Mace, The Universal Design File, NC State University, 1998; Robert G. Cooper, Winning at New Products, 5th Edn., Basic Books, 2017; Harry Brignull, Deceptive Patterns, 2023; The Institution of Engineers (India), Code of Ethics for Corporate Members; Nigel Cross, Engineering Design Methods, 4th Edn., Wiley, 2008.
